Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

- First, convert 1eV to joules. 1eV = 1.602 ÃĹ 10^-19 J
- The de Broglie wavelength is given by Î» = h / p, where h is Planck's constant and p is the momentum.
- Momentum can be calculated as p = sqrt(2 * m * E) for non-relativistic speeds, where m is the mass of the electron and E is the kinetic energy.
- The mass of an electron is approximately 9.109 ÃĹ 10^-31 kg.
- Use these values to find the wavelength.
- Planck's constant h = 6.626 ÃĹ 10^-34 Js
- We first find the momentum p and then use the de Broglie equation to find Î».

Î» = h / p

p = sqrt(2 * m * E)

E = 1.602 ÃĹ 10^-19 J, m = 9.109 ÃĹ 10^-31 kg

After calculating, the wavelength Î» will be around 1.23 ÃĹ 10^-7 cm.

</think>

<answer>1.23 ÃĹ 10^-7 cm</answer>